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Covariance

SAT scores

Background

Task 1

Task 2

Task 3

You are given data on the SAT reading and writing scores of several students from our lesson on cross tables and scatter plots

Determine if this is sample or population

Calculate the covariance of the two datasets

Plot the data on scatter plot and using your previous knowledge comment on whether there is a noticeable relationship between the two variables.

	Writing	Reading	
	344	378	19,490.16
	383	349	19,004.16
	611	503	1,179.36
	713	719	44,714.16
	536	503	234.36
Mean	517.4	490.4	84,622.20
			5
			21,156

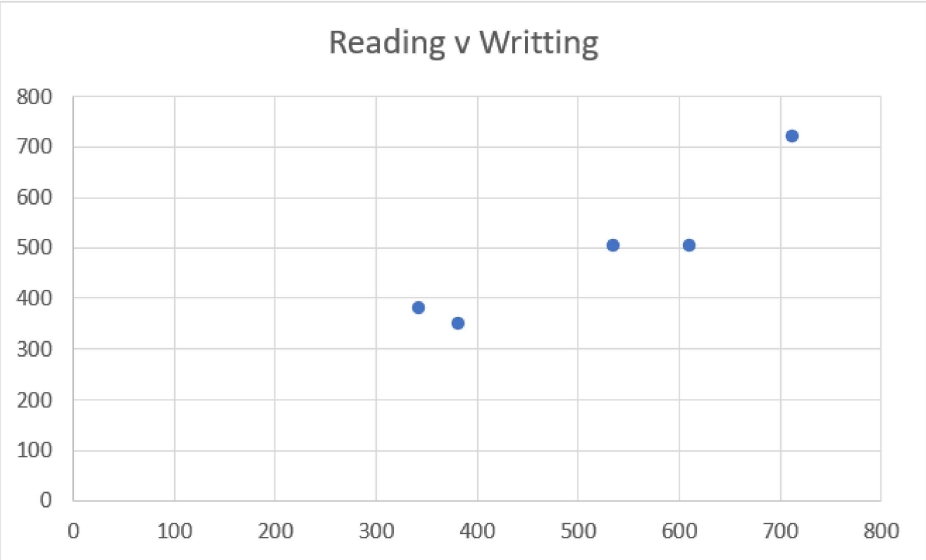
Sample size

Cov. Sample

The covariance is positive meaning the two variables move together and are therefore correlated

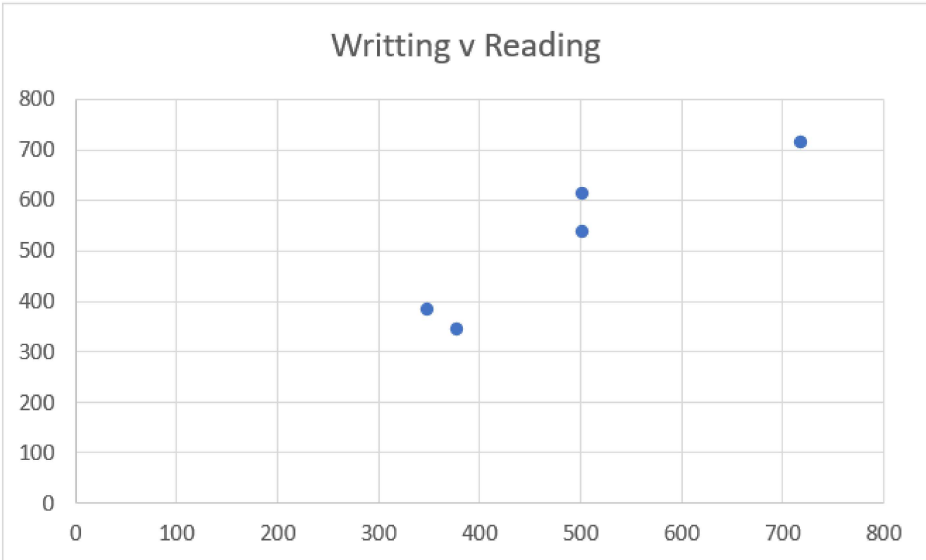
This is visually represented on the scatter plot graph

Reading v Writting



Writing	Reading
344	378
383	349
611	503
713	719
536	503

Writting v Reading



Reading	Writting
344	378
383	349
611	503
713	719
536	503